

Go-Back-N Protocol Scenarios

Assume that we have a GBN protocol with a window size of 3. We are trying to send 8 packets total (e.g., [0, 1, 2, 3, 4, 5, 6, 7]). Show both sender and receiver sides for given cases. For the receiver side, write what is the ACK number. In the sender side, show what is the packet number. Also, show the sender window each time it changes. Finally, show what happens after the timeout, e.g., Sender retransmits some packets, receiver sends ACK, show the values. Main assumptions for the question: Assume that the timeout does not occur prematurely. We do not have network congestion or additional delay.

a) All Packets Including ACKs Delivered Without Error

Sender Side:

1. Initially: Send packets 0, 1, 2.

Window: [0, 1, 2]

2. After Sending: Send packets 3, 4, 5 (after receiving ACK for packet 0).

Window: [3, 4, 5]

3. After Sending: Send packets 6, 7 (after receiving ACK for packet 1).

Window: [6, 7]

Receiver Side:

Receives packets in order: 0, 1, 2, 3, 4, 5, 6, 7.

Sends ACKs for each packet:

ACK 0 for packet 0,

ACK 1 for packet 1,

ACK 2 for packet 2,

ACK 3 for packet 3,

ACK 4 for packet 4,

ACK 5 for packet 5,

ACK 6 for packet 6,

ACK 7 for packet 7.

Final State: All packets are sent and acknowledged without any issues.

b) Only Packet #4 Loss During First Transmission

Sender Side:

1. Initially: Send packets 0, 1, 2.

Window: [0, 1, 2]

2. After Sending: Send packets 3 (ACK received for packet 0).

Window: [3, 4, 5]; Send packet 4 (lost);

Send packet 5 (after packet 3 is acknowledged).

3. After Timeout: Timeout occurs for packet 4;

Retransmit packets 4 and 5 (since it is a Go-Back-N protocol).

Window: [4, 5, 6].

4. After Sending: Send packets 6, 7 (after receiving ACK for packet 3).

Window: [6, 7].

Receiver Side:

Receives packets: 0, 1, 2, 3;

Sends ACKs:

ACK 0 for packet 0,

ACK 1 for packet 1,

ACK 2 for packet 2,

ACK 3 for packet 3;

ACK 4 is not sent (since packet 4 is lost).

After the retransmission, the receiver receives packets 4 and 5:

Sends ACK 4 after receiving packet 4;

Sends ACK 5 after receiving packet 5;

Finally, receives packets 6 and 7, sending ACK 6 and ACK 7 respectively.

Final State: Packets 0, 1, 2, 3, 4, 5, 6, and 7 are sent, with packet 4 being retransmitted successfully after a timeout.

c) Only ACK Packet #3 Loss During First Transmission

Sender Side:

1. Initially: Send packets 0, 1, 2.

Window: [0, 1, 2]

2. After Sending: Send packets 3, 4, 5 (after ACK for packet 0).

Window: [3, 4, 5]; Send packets 6, 7 (after receiving ACK for packet 1).

3. Loss of ACK #3: Host A does not receive ACK 3 for packet 3;

Sender will timeout after a predefined period for packet 3 and retransmit packets 3, 4, 5.

Window: [3, 4, 5].

4. After Sending: If the retransmission is successful, packets 3, 4, 5 are sent again, followed by packets 6 and 7 (after ACK for packet 2).

Receiver Side:

Receives packets: 0, 1, 2;

Sends ACKs:

ACK 0 for packet 0,

ACK 1 for packet 1,

ACK 2 for packet 2;

ACK 3 is lost (not received by sender).

After the retransmission:

Receives packets 3, 4, 5 successfully;

Sends ACK 3 after receiving packet 3;

Sends ACK 4 after receiving packet 4;

Sends ACK 5 after receiving packet 5;

Finally, receives packets 6 and 7, sending ACK 6 and ACK 7 respectively.

Final State: All packets are sent successfully with packet 3 being retransmitted due to the loss of ACK #3.